QA with Graph-contextualized Scientific Papers

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I. Introduction

Since the widespread use of the transformer architecture in natural language processing, there have been many advancements in input representation. For example, in the case of word embeddings and sentence embeddings, some sequence of natural language is represented as a dense vector of numbers capable of capturing some form of the text's meaning. More recently, research into multimodal models has produced methods for combining representations of multiple modalities such as images, sound, and video [7]. The primary challenge in these approaches is representing the inputs in such a way that a large language model (LLM) can utilize the information well to generate its response.

An orthogonal problem to input representation is that of context length. Many tasks require a large amount of background knowledge or context, which is not possible to provide for an LLM with a relatively small context length. Many techniques, particularly those related to retrieval augmented generation (RAG), have been devised to address this exact issue. Usually these techniques involve searching a large space of contexts, extracting important features (retrieval), and reinforcing the original prompt with these features [8].

In this project, we used a structured textual representation in order to tackle the problem of context. We tested the implementation of the Graph Contextualized framework by Rajkumar Pujari, proposed and detailed in [6]. In the framework, for each text prompt is augmented with a *graph*. The graph's nodes each correspond to some combination of *documents*, which can be in any form that can be embedded. For example, one could have nodes each correspond to a quote by some speaker. The nodes are then connected based on some structure that is to be represented, and may be useful in the task. Finally, each node can be connected to some part of the textual prompt. This framework is extremely flexible, and poses a potential solution to the issue of context length.

We used this framework for question answering (QA) of fact-based questions using a large context of scientific papers. The idea is that scientific papers form a natural graph structure in the way that they cite one another, and fact-based QA is a task that can easily benefit when the right evidence is found. Our hypothesis is twofold: (1) scientific papers provided as context will provide evidence for QA and will improve accuracy and (2) the graph structure formed by scientific papers will be useful for QA in that it connects similar ideas and relevant concepts.

Our contribution is as follows:

1) Testing Pujari's framework as a RAG-style evidence giver for QA tasks
2) Exploring whether the added graph structure contributes to the model's success in the QA task

II. Methods

A. Datasets

To confirm our hypothesis, we leveraged the OpenAlex dataset [5], which comprises of 25 million research publication records. Notably, these records include citations, forming a natural graph structure conducive to our research hypothesis. Our primary focus was on publications in the fields of Physics, Chemistry, and Biology, specifically those available in English and accessible to the public. Considering our time and resource constraints, we further filtered the dataset based on a citation percentile greater than 90%. This strategic filtering allowed us to concentrate on publications with significant impact.

During the dataset filtering process, we faced a challenge with the OpenAlex dataset: it includes metadata fields but lacks the actual content of the academic papers. To address this, we developed a script to download PDFs from the metadata's source links. We then employed the PyPDF [2] library to convert these PDFs into plain text. The script operated on a best-effort basis, disregarding any publications that could not be downloaded or converted. Ultimately, this method produced approximately 138,929 publications, amounting to 6.4 GB of data. Of these, about 42,393 publications either contained references or were referenced by others in the downloaded PDFs. We focused on these publications throughout the experiment, as they are capable of producing relational graphs.

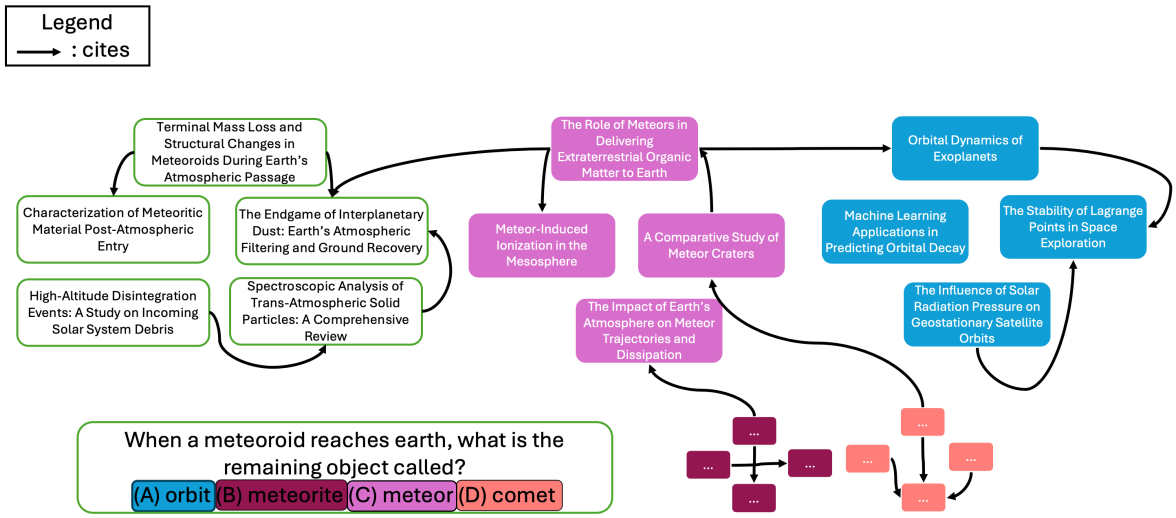


Figure 1. An example of what a context graph would look like for a given QA question. Relevant papers are obtained via ColBERT queries on five components: (1) the question itself is queried (white nodes), and (2) each answer choice is queried (colorful nodes). Papers are connected to each other based on whether they cited one another (black arrow). Finally, answer choice documents are "mapped" to the corresponding segment in the prompt.

The dataset's quality fell short of our initial expectations due to the manual parsing and the limitations of the PyPDF library. This tool often falters with non-standardized PDF formats and scanned images, which is more prevalent in older publications. A considerable preprocessing effort is essential to enhance the dataset's quality, such as stripping citation sections, handling mathematical equations and tabular data, which is feasible with additional time. It's important to recognize that converting PDFs to a clean, plain text format is a complex and significant undertaking, which on its own is a contribution.

For our approach, we leveraged the SCiQ question and answering dataset [3] in conjunction with the OpenAlex dataset for supervised fine-tuning on the 248M flan T5-base model. The SCiQ dataset comprises of 13,679 crowd-sourced science exam questions spanning Physics, Chemistry, and Biology. These questions are presented in a multiple-choice format, each with four answer options. For this dataset, we encountered issues such as duplicate distractor answers and instances where a distractor was also the correct answer, contrary to our assumptions. Consequently, we modified our training algorithm to accommodate these anomalies accordingly.

B. Training

Prior to training, we utilized RAGatouille [1], a tool that interfaces with the ColBERT [4] library, to index the 42,393 publications segmented into 512-token chunks, keeping the references of each publication in the metadata of each index record. ColBERT is an advanced retrieval model that leverages BERT for generating an embedding index from a document collection. It also facilitates an efficient BERT-based search across this indexed corpus using embedding similarity and 'late interaction'.

Our approach involves training on the questions from SCiQ and incorporating constructed citation graphs from the OpenAlex dataset at specific prompt tokens. We retrieve the top 50 similar publication segments using RAGatouille for the question and the top 4 for each answer choices. Finally, we connect all these papers in a graph based on their citation relationship, and attach the papers to the keywords in the prompt. Figure 1 demonstrates this process.

The key idea is that the papers will form clusters based on their topics, which will help the language model answers questions about those topics. This process is repeated for each question in the SCiQ dataset, and the prompt is augmented with the contextual graph before fine-tuning on the 248M flan T5-base model.

C. Baselines

Our metric is the multiple choice accuracy on the validation and test partition of the SciQ dataset. We expect the model to output one of A, B, C or D which is then matched with the true label provided by SciQ. We compared our approach to three baseline models using the accuracy score from the performance of each models on the SCiQ dataset. The three baseline approaches include:

a. Supervised fine tuning on the SciQ dataset only with no graphs. This serves as a low bar to surpass, since in this case, the model has no access to any additional information (papers).

Table I
EXPERIMENT RESULTS

Model	Test Accuracy (%)	# of Train Epochs
Random	25.0	-
Baseline (a)	71.1	100
Baseline (b)	27.0	1
Baseline (c)	74.3	5
Ours	28.8	1

b. Supervised fine tuning on the SciQ dataset with the same papers in the graph, but where the graph is fully connected.
c. Supervised fine tuning on the SciQ dataset with as much of the top relevant publication segments of the question and answer prepended to the prompt that fits within 512 tokens.

Baseline (a) serves to show that adding the additional information is helpful (since it has none). Baseline (b) serves to show that the structure induced by the citations is helpful for the model in weighing importance/finding important features of the papers. Baseline (c) serves to show that the graph and its information are at all useful, compared with simply attaching relevant information to the prompt.

III. RESULTS

As can be seen in the results, we had severe issues with compute and time required to train. While non-graph baselines (a) and (c) could be trained in batches and did not need a large set of documents in memory, baseline (b) and our experiment took much, much longer. One epoch ($\sim$11,000 training examples) took about 2 days (each) and required > 20 GB memory on one GPU. Because of this, we were only able to train for one epoch for baseline (b) and our own framework.

It should be noted that the SciQ questions are always multiple choice and have exactly four options, so guessing randomly yields an accuracy of 25%. As can be seen, baseline C outperformed baseline A as expected since baseline C incorporates more information from relevant segments. The graph models perform significantly worse (nearly random), though it is very difficult to compare them with only one epoch of training.

IV. ANALYSIS

Because of the constraints of training time and GPU memory, it is very difficult to conduct an in-depth comparison of the results. This is because our framework was only trained for one epoch, whereas two of the baselines were trained for 5, and 100 respectively. It is even difficult to compare the number of epochs directly, because baselines (a) and (c) contained at most 512 tokens in each input, whereas baseline (b) and our framework included 66 $(50+4*4)$ paper chunks, each with up to 512 tokens. This means each input was up to $512*66 =\sim 34,000$ tokens. Because of the difference in size of the input, we hypothesize that it would take many more training iterations to reach a high accuracy, since the model must learn which features are important and which to ignore in a much bigger input space.

Despite these issues, we can still conclude a few things. Firstly, the difference between baselines (a) and (c) demonstrates that the context provided by the scientific papers is in fact helpful for this QA task. This is important, as it shows that in principal, this task and dataset are compatible with proving our hypothesis about graph-contextualized prompt augmentation.

Secondly, we show that the graphical models are in fact able to learn, albeit very little within the one training epoch. With a test partition of 400, we can be quite confident that the model has learned at least some useful features for answering the questions. This is true for both the fully connected graph baseline (no citation information) at 27.0% and for our framework at 28.8%.

Finally, we confirm the power of the ColBERT system in performing useful RAG for our task. We see that this is the case both with the improvement from baseline (a) to (c), and also upon manual inspection of the retrieved documents.

V. FUTURE WORK

There are several direct ways to continue this line of work. The first and most obvious is to continue training the models for many more epochs. We realized that the main reason that training the graph model takes so long is not the actual training step, but is in fact the creation of a graph for each question. This comes even after pre-retrieval using ColBERT of all relevant documents. We propose solving this issue by pre-pickling all components of the graph and saving them to disk so that for each problem, only backpropagation needs to be done. We estimate that this will speed up training time by about 20x.

The next step would be reducing the chunk size for the indexed documents. Due to GPU/timing constraints, we were only able to segment each paper into chunks of size 512. While we hypothesize that this is still much better than having the whole paper as a node in the graph, it is likely that there is still very much irrelevant data among those tokens. We conjecture that it would be much better to segment the documents into small chunks, providing only the most relevant content to the question/answer choice. Then we could also include more documents when constructing the graph without running into memory limitations. The only drawback of smaller document size is that as each node gets smaller, the meaning of a

citation edge between two nodes means less and less, as the reason for one paper's concept being related to another becomes less and less likely to be in that specific chunk. Despite this, we think it would be potentially very helpful for the model when it chooses its answer.

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